

Master of Science
in

Knowledge and Technology Management

Managing Knowledge (ISE542)



THE HONG KONG
POLYTECHNIC UNIVERSITY

香港理工大學

Department of Industrial
& Systems Engineering

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Subject Coordinator

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Learning Objectives

The aims of the subject are to:

1. Introduce the students the foundations of KM.
2. Provide the students with an insight and introductory working knowledge to allow them to effectively apply KM in organizations to achieve their objectives and lead and promulgate KM efforts for business effectiveness and success.
3. Equip the students with the practitioner understanding and proficiency to initiate, assess, operate, disseminate, and manage KM practices, projects, programs, and other KM efforts with an enterprise.



Outline Syllabus

Knowledge Management Essentials

Evolution of knowledge, theory and concepts of KM, type of knowledge, nature of knowledge work

Managing Knowledge Processes

Various knowledge processes and their applications, methods and tools for managing knowledge processes

Knowledge Auditing

Role and importance of knowledge audit, knowledge audit process and practices , methods and tools for conducting knowledge auditing, analysis and reporting the results of a knowledge audit

Outline Syllabus

Knowledge Management Related Strategies

KM strategy and how to manage culture aspects, and initiatives of KM, People-centric and IT-based KM from business perspectives based on understanding that business performance results from knowledgeable (competent), motivated, and accountable human actions, in part supported by IT capabilities.

Knowledge Management Practices

KM practices in different industries, the management of KM projects, ethical issues in KM, case studies



Teaching Plan

Semester Week No. (Day)	Activities	Staff	Venue
1 (3/9)	Subject Briefing Lesson 1: The Evolution of Knowledge - The evolution of the concept of knowledge - Defining knowledge - Differentiate various types of knowledge - To explain the way in which knowledge can be converted (Nonaka's cycle of knowledge conversion) - Meta-level of knowledge.	Benny Cheung & Eric Tsui Benny Cheung	V312
2 (10/9)	Lesson 2: What is Knowledge Management? - The concept of Knowledge Management - Nature of knowledge work and knowledge workers - Defining Knowledge Management - Reviewing how knowledge can be processed - To realize the multi-disciplinary nature of KM - To identify the common questions, success factors and benefits of Knowledge Management	Eric Tsui	V312
3 (17/9)	Lesson 3: Knowledge Processes (1): Knowledge Acquisition, Knowledge Generation and Innovation - What are knowledge processes? - Knowledge acquisition, knowledge generation and innovation processes and practices - Differentiate between innovation, knowledge acquisition and knowledge generation - Knowledge validation as a way to determine the quality of knowledge	Benny Cheung	V312
4 (24/9)	Lesson 4: Knowledge processes (2): Knowledge Transfer, Sharing and Retention - Familiar with knowledge transfer, sharing and retention - Knowledge transfer methods and how to support effective knowledge sharing - Develop culture for knowledge transfer - Various techniques for knowledge retention such as after-action review, etc.	Benny Cheung	V312



Teaching Plan

5 (2/10) Make-up class 6:30 pm – 9:30 pm	Lesson 5: Knowledge Management Systems and Tools - Role of IT in KM - Evolution of KM systems over the past 3 decades - Search engines and technologies - Value Creation by KMS - AI-enhanced KMS - Case Studies	Eric Tsui	Online
6 (8/10)	Lesson 6: Knowledge Management Related Strategies - Business strategy drives KM. - Differentiate KM approaches to increase customer intimacy. - Methods in pursuing product and services leadership in KM. - Review existing KM implementation models Workshop: Part I – KM Strategy Briefing	Benny Cheung	V312
7 (15/10)	Lesson 7: Knowledge Auditing – Methods and Tools - The role and importance of a knowledge audit - The knowledge audit process. - The benefits of conducting a knowledge audit and the hurdles involved. - Various types of knowledge audit methods - Tools for carrying out a knowledge audit - KM related planning	Benny Cheung	V312
8 (22/10) 9:30 am – 12:30 pm	Lesson 8: Knowledge Auditing – Analysis and Reporting - Various methods for analyzing and reporting the findings of a knowledge audit such as Knowledge diagnostics, benefit analysis, knowledge network analysis, SWOT and social network analysis. - Use the findings of the knowledge audit findings for strategic planning. - How to implement KM initiatives and take action on the suggestions that result from a knowledge audit. - Alignment of the knowledge audit findings with business strategies.	Benny Cheung	V312
	Workshop 2: Knowledge Auditing Tools and Practices		



Teaching Plan

9 (29/10)	Lesson 9: Managing Knowledge in Different Industries - Identify the driver for adopting KM in various industries. - Differentiate and explain various KM practices used in different industries.		Eric Tsui	V312
10 (5/11)	Lesson 10: Managing KM projects and applying KM to Project Management - Planning for a KM project - Challenges and Risks in KM projects - KM team compositions - Measuring success of KM projects - How KM can benefit Project Management? - AI-enhanced Project Management support		Eric Tsui	V312
11 (12/11)	Lesson 11: Managing and Culture and Taxonomy for KM Implementation - What an enterprise culture is? - The importance of culture aspects for KM. -Applying methods for fostering knowledge-supportive mentality and culture. - Derive incentive and culture drivers - Roles of Taxonomy and Its Applications in KM - Method and tools for building and maintaining Taxonomy		Benny Cheung	V312
12 (19/11)	Workshop: Part II – KM Strategy Presentation (Session A) – V312	Workshop: Part II – KM Strategy Presentation (Session B) – V302	Benny Cheung and Eric Tsui	V312& V302
13 (26/11)	Open Book Test		Benny Cheung	V322

Learning Outcome

Upon completion of the subject, students will be able to:

- a. understand the concepts and ethical issues in KM
- b. select and devise KM strategies, programmes and actions to provide effective business support
- c. conduct knowledge audit and use of various knowledge audit methods for analyzing and reporting findings for KM related planning
- d. To conduct a feasibility study on the implementation of knowledge management projects in an organization



Table 1 Assessment Methods and Criteria

Activities	Marking method	%	Submission
Personal Learning Environment and Network (PLE&N)	Each student is required to set up a Personal Learning Environment and Network (PLE&N) which is an on-line chat room. The students are required to contribute to their discussions and learning in this environment during the semester and beyond.	10	Assessment will be based on evidence of consistent and constructive contributions and discussions throughout the semester. The participants are expected to demonstrate consistent effort throughout the semester. Feedback will be given during the semester.
Individual Assignment	Each student is required to complete an Individual Assignment. The assignment questions <u>aims</u> to facilitate the students to reflect what they learn in classes and how to apply the theory in the real life examples.	20	On-line submission via Blackboard. Information on how to submit assignments will be provided in due course.
Workshop Assignment	Presentation: The performance of the <u>participant's presentation</u> in the KM Strategy workshop will be individually assessed.	10	The participants are required to make a presentation for their KM strategy.
	Written group report of <u>KM strategy</u> workshop	15	The participants are required to submit a group written report for KM strategy workshop.
Open Book Test	Three-hour open book test	45	The open book test will be held at the end of the semester.

Assessment Plan



Department of Industrial & Systems Engineering

The top screenshot displays the L@PU LEARN@PolyU (理學網) login page. It features the university's logo and name in both English and Chinese. A red banner reads "Welcome to LEARN@PolyU (理學網)". Below this, a login section asks "Have an account?" and prompts the user to enter their "Username:" and "Password:". A "Login" button is provided, along with a link for "Forgot Your Password?". To the right, a "Quick Link" section offers direct access to "Subject Registration" and "Blackboard Homepage".

The bottom screenshot shows the "Managing Knowledge" interface within the L@PU system. The user is logged in as "Chi Fai CHEUNG". The interface includes a top navigation bar with links to "Courses", "Community", "Content Collection", "Library", and "SPECIAL ePortfolio". A sidebar menu on the left lists various resources like "Announcements", "Home Page", "Introduction", "Experts", "Contents Materials", "Bulletin Boards", "Messages", "Calendar", "Download", "Assignment", and "Chat". The main content area, titled "Home Page", contains buttons for "Build Content", "Assessments", "Tools", and "Publisher Content", along with a "Discover Content" link. Below these, a list of resources is shown, including "ISE542 - Managing Knowledge", "Introduction", and "Content Materials".